Hypoxia-Inducible Factor 1-Alpha Orchestrates VEGF-Mediated Angiogenesis in Ischemic Tissue

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Page 1 — Abstract & Introduction

Tissue ischemia induces oxygen deprivation, triggering compensatory vascular remodeling. Hypoxia inhibits prolyl hydroxylase domain enzymes, preventing the degradation of key transcription factors.

Page 2 — Mechanisms

Inhibition of prolyl hydroxylases by low oxygen stabilizes HIF-1alpha by blocking its binding to von Hippel-Lindau protein.

Page 3 — Results: Gene Expression

Stabilized HIF-1alpha translocates to the nucleus and upregulates VEGF expression by binding the hypoxia response element.

Page 4 — Results: Endothelial Response

Secreted VEGF binds VEGF receptor 2 on endothelial cells, activating downstream signaling to stimulate angiogenesis and tube formation.

Page 5 — Discussion

Consequently, hypoxia promotes angiogenesis via sequential stabilization of HIF-1alpha and upregulation of VEGF. Normoxic conditions failed to stimulate VEGF expression.